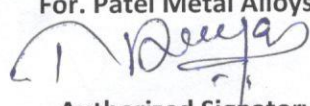




Plot No. 4300-4301 G.I.D.C Phase-III Dared, Jamnagar-361004 (Gujarat) Ph:0288-2730060
e-mail. patelmetalalloys@gmail.com GST No.24AAVFP6629H1Z9
Manufacture All Kind of Brass Rods, Section, Hollow Rods.

TEST CERTIFICATE

Customer:		MTC No. 3212	Date: 16-12-2023		
METAL MECH ENGINEERING		Invoice No. PMA/1409	Date: 16-12-2023		
JAMNAGAR		P.O. No. 2EX/0460/23/24	Date:		
		Qty. (Kgs.).			
We hereby certify that above material conforms to the following Chemical composition & Physical properties.					
PARTICULARS	As Required by Specification	Specification of Supplied Material			
I. CHEMICAL COMPOSITION	EN 12164 CW614N CUZN39PB3	Size	Size	Size	Size
		55 X 36 MM ✓ HEX HOLLOW	65 X 47 MM ✓ HEX HOLLOW	50 X 38 MM ✓ HEX HOLLOW	
		Heat No.	Heat No.	Heat No.	Heat No.
		C-254 ✓	C-251 ✓	C-252 ✓	
		290.250 Kgs.	302.950 Kgs.	183.600 Kgs.	
1. COPPER (Cu) %	57.00 – 59.00	57.68	57.70	57.52	
2. LEAD (Pb) %	2.50-3.50	2.84	2.80	2.81	
3. IRON (Fe) %	0.30 Max	0.25	0.25	0.27	
4. ZINC (Zn) %	Remainder	Remaining	Remaining	Remaining	
5. ALUMINIUM (Al) %	0.05 Max	0.04	0.02	0.02	
6. MANGANESE (Mn) %	-	-	-	-	
7. NICKLE (Ni) %	0.30 Max	0.22	0.28	0.23	
8. TIN (Sn) %	0.30 Max	0.29	0.27	0.29	
9. SILICON (Si) %	-	-	-	-	
10. PHOSPHORUS (P) %	-	-	-	-	
11. SULPHUR (S) %	-	-	-	-	
12. TOTAL IMPURITIES %	0.20 Max	0.07	0.07	0.08	
II. PHYSICAL PROPERTIES					
1. UTS – Min (Mpa)	430 MIN	439	439	439	
2. ELONGATION %	20 MIN	26	26	26	
3. HARDNESS (HV5)	-	-	-	-	
4. IZOD	-	-	-	-	
5. 0.2% PROOF STRESS	350 MAX-	342	342	342	
III. Mercurous Nitrate Test					
6. CONDUCTIVITY	-	-	-	-	
For. Patel Metal Alloys  Authorized Signatory					

